

ML Practice Projects — Roadmap

Consolidation phase after Geron Chapters 2-4 · 3 Classification + 3 Regression · updated 2026-06-26

Purpose

After finishing Geron Chapters 2-4, the goal is to consolidate by building practice projects before moving deeper into the book. Each project is treated as a permanent **reference notebook** and must add a NEW skill beyond the previous one. Classification and regression are interleaved so the evaluation concepts keep contrasting.

Completed

P1 - UCI Heart Disease (Cleveland) · binary classification · **DONE**

Path: classification/project_4_UCI_Heart_Disease/Heart_Disease_P1_Clinical_Prediction.ipynb

Now a rich template covering: full EDA (skew, outliers, chi-square, ANOVA, VIF, Spearman + effect sizes), a custom **AgeGroupImputer** (grouped imputation, Titanic-style), ColumnTransformer pipeline, a 12-model bake-off, GridSearch tuning, and a **Rigor & Honesty** section (demographics-only screening-vs-diagnosis test, RepeatedStratifiedKFold distribution, bake-off error bars, Youden-J clinical threshold, learning + validation curves). Ends with a QUICK-RECALL cheat-sheet.

The 6-Project Roadmap

#	Project	Type	New skills beyond P1
1	Telco Customer Churn	Binary (imbalanced)	SMOTE / class_weight, PR-AUC, calibration, cost-based threshold, high-cardinality encoding
2	Ames Housing	Regression	80 features, ordinal encoding, heavy missingness, log-transform target, feature selection
3	Fetal Health (CTG)	Multiclass (medical)	macro-F1, K x K confusion matrix, one-vs-rest ROC, multiclass + imbalance together
4	Medical Insurance Cost	Regression	interaction terms (smoker x age x BMI), small clean data, interpretation focus
5	Covertypes	Multiclass (big data)	scale (581k rows), sampling, efficiency / training-time tradeoffs
6	Energy Efficiency	Regression	multi-output (heating + cooling load), tiny dataset, near-deterministic signal

Suggested Order (interleaved)

Churn → Ames → CTG → Insurance → Covertypes → Energy

Difficulty ramps gently: Churn and Insurance reuse the P1 pipeline; Ames (feature breadth) and Covertypes (data scale) are the heavy ones. Never two of the same task in a row.

Why These Projects - the Skills Gap After P1

P1 was binary, balanced, small, and clean. It never forced these skills, so the roadmap targets them:

Missing skill	Covered by
Severe class imbalance + resampling (SMOTE)	Churn, CTG
PR curve / PR-AUC (honest metric for rare positives)	Churn
Probability calibration	Churn
Cost-based decision threshold	Churn, CTG
Multiclass (macro-F1, K x K confusion, OvR ROC)	CTG, Covertypes
High-cardinality categorical encoding	Churn
Large-scale data (sampling, efficiency)	Covertypes
Feature-rich regression + regularization	Ames
Interaction terms + interpretation	Insurance
Multi-output regression	Energy Efficiency

The Reusable Template (every project follows this)

1. EDA + stats (skew, outliers, chi-square/ANOVA, correlation + effect size, VIF) 2. Split FIRST 3. Impute → scale/encode inside a Pipeline (no leakage) 4. Dummy baseline + multi-model bake-off (CV) 5. Hyperparameter tuning 6. Lock-box test evaluation 7. Rigor & Honesty checks 8. Notes + quick-recall

The 5 rules that never change: (1) split before you learn anything; (2) beat the dumb baseline; (3) right metric, not just accuracy; (4) significance is not strength - pair p-values with effect size; (5) distrust scores that are too good.

Next up: Telco Customer Churn

Data: Kaggle [blastchar/telco-customer-churn](#) (single CSV, ~7,000 rows, ~27% churn). New muscle: the imbalance toolkit - PR-AUC over ROC, SMOTE vs class_weight, calibration curves, and a cost-based threshold.

Known gotcha (catch it in EDA, do not pre-clean): TotalCharges is stored as text with blank strings for tenure-0 customers.